Schema-Driven Framework for Data Submission, Integration, and Dissemination in the Pan-Canadian Genome Library

Pan-Canadian Genome Library
Bibliothèque génomique pancanadienne

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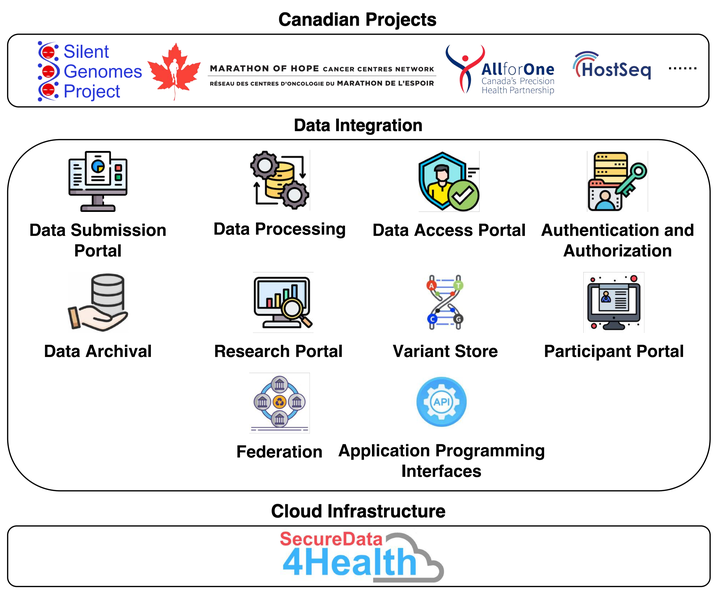
Introduction

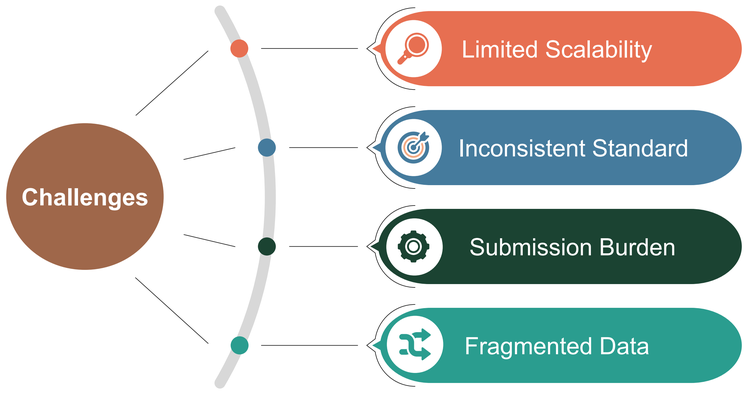
The Pan-Canadian Genome Library (PCGL) is an open-source and open-science initiative:

- ❖ Build upon Canadian-made foundational components and datasets
- ❖ Utilize international standards
- ❖ Unify Canada's human genome sequencing efforts
- ❖ Set out a federated data management system

PCGL Infrastructure

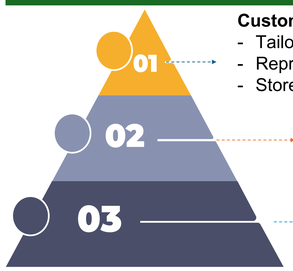


Key Data Schema Challenges



Approach

Three-Tiered Schema Structure

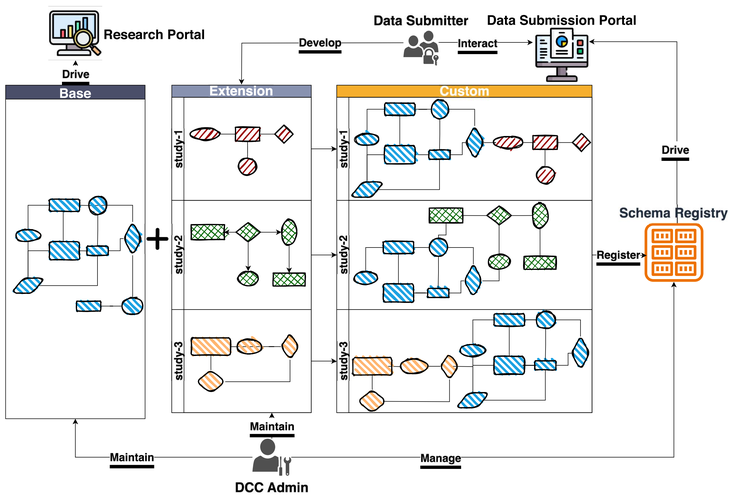


Custom
- Tailored to individual studies
- Represents the complete schema used by studies
- Stored, managed and versioned by Schema Registry

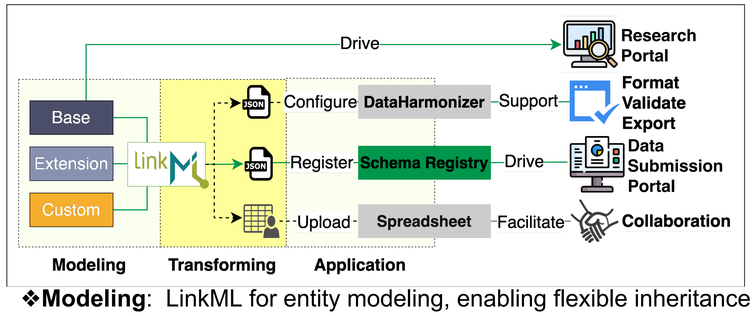
Extension
- Study-specific fields
- Collaboratively developed by studies and PCGL
- Meet precise needs without affecting the Base

Base
- Standardized, and ontology-driven fields
- Defined and maintained by PCGL
- Powers Research Data Portal to ensure consistency and interoperability across datasets

Schema-Driven Framework



Implementation

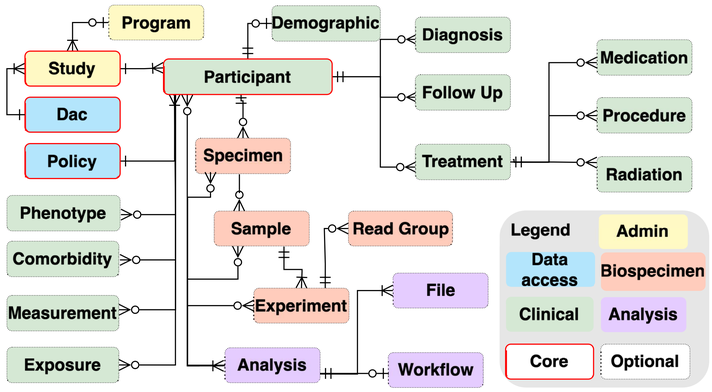


- ❖**Modeling**: LinkML for entity modeling, enabling flexible inheritance and ontology mapping
- ❖**Transforming:** Customized scripts
- ❖**Application:**
  - ➢ Schema registry is powered by overture
  - ➢ Exploring the potential use of DataHarmonizer

Scan here for data dictionary

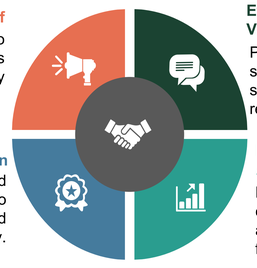
Results

Base Schema Design



PCGL Base Schema V2.0

Framework Features



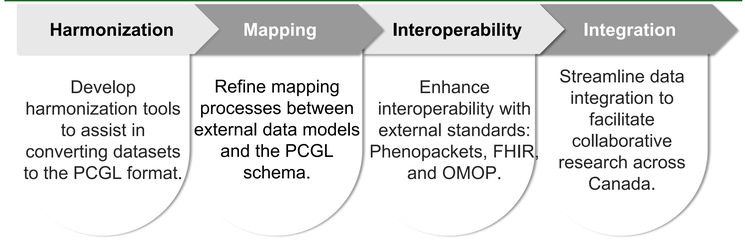
Scalable & Future Proof
Use extensible schema to support evolving research needs without breaking compatibility

Efficient Submission & Validation
Provide automated validation, structured templates and study-specific extensions to reduce submission burden.

Standardization
Implement ontology driven and cross-platform annotation to ensure data consistency and interoperability.

Data Integration & Accessibility
Enable proper linkage between clinical, genomic, and analytical data to minimize data fragmentation.

Next Steps



Summary

- ❖ **Flexible & Modular**: Design three-tiered schema to support diverse data submissions.
- ❖ **Governed & Interoperable**: Maintain schema versioning, validation, and compliance, with integration into external standards like Phenopackets, FHIR, and OMOP.
- ❖ **Streamlined & Accessible**: Enable efficient data submission and validation, proper linkage between clinical, genomic, and analytical data to minimize data fragmentation.

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CIHR IRSC
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